PROBLEM 22

Particle on a Curve

Original statement · Problems.pdf, p. 10

1D Curve Constraint

A heavy point particle P of mass m is constrained to move without friction along the curve:

$$y = a\left(1 - \cos\frac{x}{a}\right) \tag{4}$$

in the range $-\pi a \leq x \leq \pi a$, where a is a positive constant. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

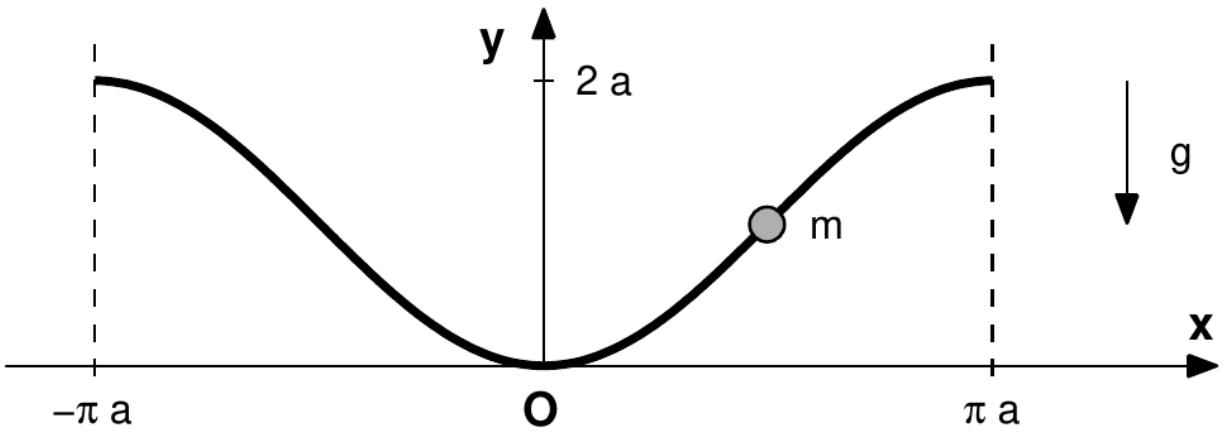


Figure: Heavy particle m on the curve y = a(1 - cos(x/a)) between x = -pi a and x = pi a, gravity g downward (Problems.pdf, p. 10).

(a) Reduce the problem to integrals.

(b) If $v_0 = \sqrt{ga}$, determine the minimum and maximum value of x along the motion.

(c) What should be the minimum value of v_0 in order to reach $x = \pi a$?

Let's assume now that the curve containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega \mathbf{j}$.

(d) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Determine the inertia force that acts on the particle in this moving reference frame as a function of m, Ω, x_1 and $\dot{x}_1$, where x_1 represents the position coordinate of P along OX_1.

(e) Reduce the problem to quadratures and determine the value of the reaction force.

(f) Find out a trascendental [sic] equation whose solutions are the equilibrium positions of the point particle in $OX_1Y_1Z_1$.

(g) If Ω were a function of time, would it be possible to use the energy equation in the same way as before? Justify your answer.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a particle P of mass m moves without friction along the smooth planar curve

$$y = a\left(1 - \cos\frac{x}{a}\right), \qquad x \in [-\pi a, \pi a]$$

in a vertical plane under uniform downward gravity $\mathbf{g} = -g\,\mathbf{j}$. In parts (a)–(c) the curve is fixed; in parts (d)–(g) it turns about the vertical axis OY with constant angular velocity $\boldsymbol{\omega} = \Omega\,\mathbf{j}$.

- **Hypotheses:** the frame $OXYZ$ with OY vertical upwards is inertial; the curve is smooth, so its reaction has no component along the curve; in (d)–(g) the rotation is imposed (whatever torque is needed is supplied).

- **Initial state ($t = 0$):** $x(0) = 0 \implies y(0) = a(1 - \cos 0) = 0$, with initial velocity $\mathbf{v}_0 = v_0\,\mathbf{i}$ ($v_0 > 0$).
- **Fixed (inertial) frame:** $S_0 = OXYZ = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, basis $\mathcal{B}_0$, with $\mathbf{j}$ vertical upwards and the curve in the plane OXY at $t = 0$.
- **Moving frame (parts d to g):** $S_1 = OX_1Y_1Z_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the curve: OX_1 in the plane of the curve, $OY_1 \equiv OY$ (the axis of rotation, so $\mathbf{j}_1 = \mathbf{j}$) and OZ_1 normal to the plane of the curve. S_1 coincides with S_0 at $t = 0$.
- **Bases attached to the curve:** the unit tangent $\mathbf{e}_t$ (in the sense of increasing abscissa) and the in-plane unit normal $\mathbf{n}$ at P; the third vector is the normal to the plane of the curve ($\mathbf{k}$, or $\mathbf{k}_1$ when the curve turns). They are deduced in the next section.
- **Coordinates and degrees of freedom:** the particle has 3 coordinates; the curve imposes two holonomic constraints, $z = 0$ (plane of the curve) and $y = a(1 - \cos(x/a))$ (in S_1: $z_1 = 0$ and $y_1 = a(1 - \cos(x_1/a))$). CDOF $= 3 - 2 = 1$, parametrized by the abscissa $x(t)$ in (a)–(c) and by $x_1(t)$, the position of P along OX_1, in (d)–(g). In S_0 the constraints of the rotating curve are rheonomic (they turn with Ωt), but the count is the same.

Symbol	Meaning	SI unit
m	mass of P	kg
a	length scale of the curve	m
g	gravitational acceleration	m/s^2
v_0	initial speed at O	m/s
Ω	angular velocity of the curve (constant, parts d–f)	rad/s
x, x_1	abscissa of P (fixed curve, rotating curve)	m

Change of Basis and Position Vector

Position vector and the slope triangle

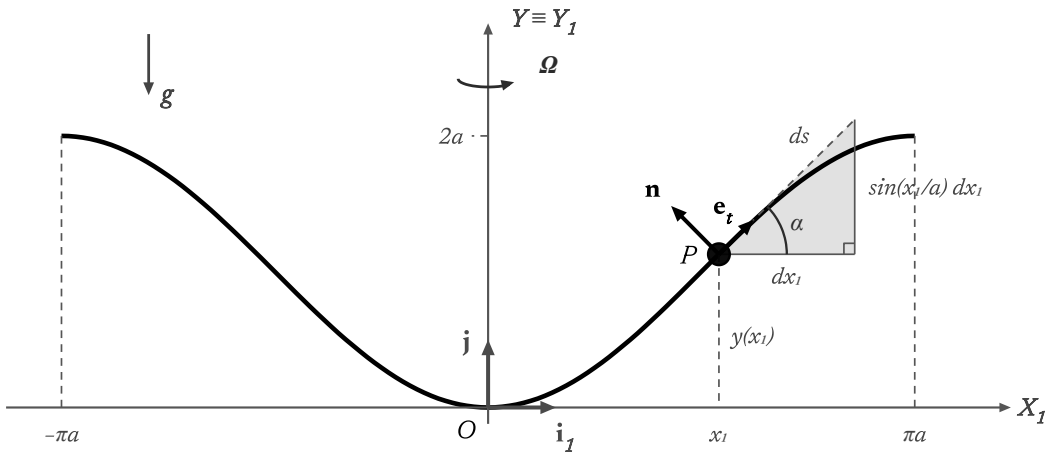


Figure 1. Plane of the curve (fixed in S_0 for (a)–(c), fixed in S_1 for (d)–(g)). The dashed lines are the components x_1 and $y(x_1) = a(1-\cos(x_1/a))$ of the position. In the shaded slope triangle the leg adjacent to the tangent angle α is dx_1, the opposite leg is $dy = \sin(x_1/a)\,dx_1$ and the hypotenuse is the arc element ds. $\mathbf{i}_1, \mathbf{j}$ in blue, $\mathbf{e}_t, \mathbf{n}$ in orange.

The plane of the curve is drawn with the horizontal axis to the right and $\mathbf{j}$ up. For the fixed curve the axis is OX with $\mathbf{i}$; for the rotating curve the same drawing holds in S_1, with OX_1, $\mathbf{i}_1$ and x_1. The position of P is the sum of its two dashed components, the abscissa along the horizontal axis and the height given by the equation of the curve:

$$\mathbf{r}^P = x\,\mathbf{i} + a\left(1-\cos\frac{x}{a}\right)\mathbf{j} \quad \text{(fixed curve)}, \qquad \mathbf{r}_1^P = x_1\,\mathbf{i}_1 + a\left(1-\cos\frac{x_1}{a}\right)\mathbf{j} \quad (\text{in } S_1)$$

Checks: $x = 0$ gives $\mathbf{r}^P = \mathbf{0}$ (the initial position O); $x = \pm\pi a$ gives the height $a(1-\cos(\pm\pi)) = a(1-(-1)) = 2a$, the two crests.

Slope of the curve. By the chain rule, $\frac{d}{dx}\cos\frac{x}{a} = -\sin\frac{x}{a}\cdot\frac{d}{dx}\left(\frac{x}{a}\right) = -\frac{1}{a}\sin\frac{x}{a}$, so

$$\frac{dy}{dx} = \frac{d}{dx}\left[a\left(1-\cos\frac{x}{a}\right)\right] = a\cdot\frac{1}{a}\sin\frac{x}{a} = \sin\left(\frac{x}{a}\right)$$

Slope triangle. Moving a distance dx along the curve, the height changes by $dy = \sin(x/a)\,dx$. In the shaded right triangle of the figure the leg adjacent to the angle α of the tangent is dx, the leg opposite to α is $dy = \sin(x/a)\,dx$ and the hypotenuse is the arc element ds. Pythagoras' theorem gives

$$ds^2 = dx^2 + dy^2 = dx^2 + \sin^2\left(\frac{x}{a}\right)dx^2 = \left[1 + \sin^2\left(\frac{x}{a}\right)\right]dx^2 \;\Rightarrow\; ds = \sqrt{1 + \sin^2\left(\frac{x}{a}\right)}\,dx$$

and, from the definitions of tangent, cosine and sine in the same triangle,

$$\tan\alpha = \frac{dy}{dx} = \sin\frac{x}{a}, \qquad \cos\alpha = \frac{dx}{ds} = \frac{1}{\sqrt{1 + \sin^2(x/a)}}, \qquad \sin\alpha = \frac{dy}{ds} = \frac{\sin(x/a)}{\sqrt{1 + \sin^2(x/a)}}$$

Unit tangent and in-plane normal. The unit tangent points along the hypotenuse, at the angle α from the horizontal axis: $\mathbf{e}_t = \cos\alpha\,\mathbf{i} + \sin\alpha\,\mathbf{j}$. The plane is seen with $\mathbf{k} = \mathbf{i}\times\mathbf{j}$ out of the paper ($\odot$), so a counter-clockwise turn of $+90^\circ$ in the drawing is a positive rotation about $\mathbf{k}$; it takes $(\cos\alpha, \sin\alpha)$ to $(-\sin\alpha, \cos\alpha)$ and gives the in-plane normal $\mathbf{n}$:

$$\mathbf{e}_t = \frac{\mathbf{i} + \sin(x/a)\,\mathbf{j}}{\sqrt{1 + \sin^2(x/a)}}, \qquad \mathbf{n} = \frac{-\sin(x/a)\,\mathbf{i} + \mathbf{j}}{\sqrt{1 + \sin^2(x/a)}}$$

Checks: $|\mathbf{e}_t|^2 = |\mathbf{n}|^2 = \dfrac{1+\sin^2(x/a)}{1+\sin^2(x/a)} = 1$; $\mathbf{e}_t \cdot \mathbf{n} = \dfrac{-\sin(x/a)+\sin(x/a)}{1+\sin^2(x/a)} = 0$; $\mathbf{e}_t \times \mathbf{n} = \dfrac{1\cdot 1\,(\mathbf{i}\times\mathbf{j}) - \sin^2(x/a)\,(\mathbf{j}\times\mathbf{i})}{1+\sin^2(x/a)} = \mathbf{k}$, so $\{\mathbf{e}_t, \mathbf{n}, \mathbf{k}\}$ is right-handed. In S_1 the same formulas hold with x_1, $\mathbf{i}_1$ and $\mathbf{k}_1$.

The basis $\mathcal{B}_1$ seen from above (parts d to g)

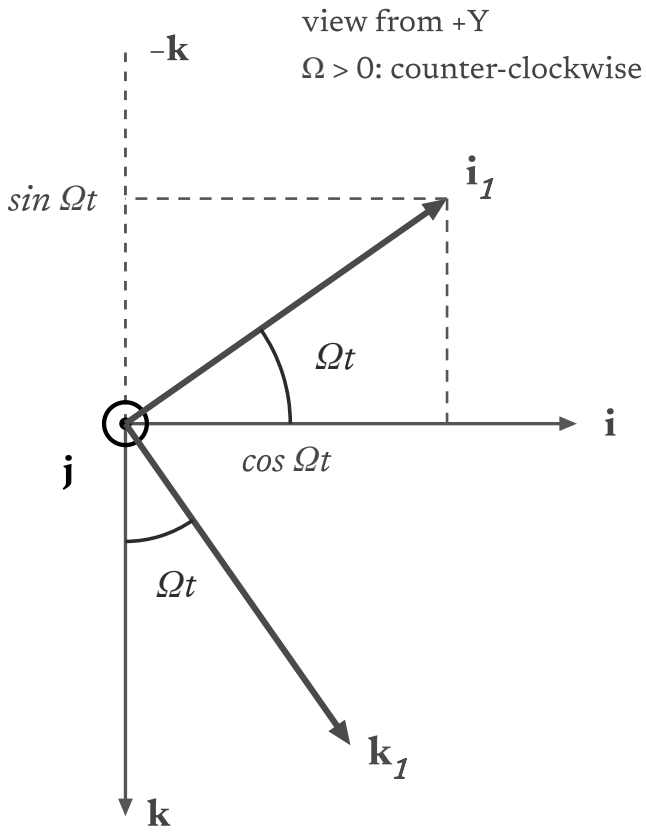


Figure 2. View from $+Y$, with $\mathbf{j}$ out of the paper ($\odot$): $\mathbf{k} = \mathbf{i}\times\mathbf{j}$ points down the page. The plane of the curve contains $\mathbf{i}_1$ and $\mathbf{j}$ and turns counter-clockwise by Ωt: the components of $\mathbf{i}_1$ are $\cos\Omega t$ along $\mathbf{i}$ and $\sin\Omega t$ along $-\mathbf{k}$ (up the page), hence the minus sign in $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}$.

The curve turns about $\mathbf{j}$, so the rotation is drawn looking down the axis, from $+Y$. In that view $\mathbf{j}$ points out of the paper, towards the reader ($\odot$). With $\mathbf{i}$ to the right, the right-hand rule $\mathbf{k} = \mathbf{i}\times\mathbf{j}$ (fingers from $\mathbf{i}$, to the right, towards $\mathbf{j}$, towards the reader) puts the thumb, $\mathbf{k}$, down the page; $-\mathbf{k}$ points up the page.

With $\boldsymbol{\omega} = \Omega\,\mathbf{j}$ and $\Omega > 0$ the plane of the curve turns counter-clockwise in this view (thumb along $\mathbf{j}$, towards the reader). After a time t the axis $\mathbf{i}_1$ has turned the angle Ωt from $\mathbf{i}$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the dashed right triangle. The leg adjacent to Ωt lies along $\mathbf{i}$ and measures $\cos\Omega t$; the leg opposite to Ωt points up the page, the direction $-\mathbf{k}$, and measures $\sin\Omega t$. Hence $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} + \sin\Omega t\,(-\mathbf{k})$: **the minus sign comes from $\mathbf{k}$ pointing down the page in this view.**
- $\mathbf{k}_1$ starts on $\mathbf{k}$ (down the page) and turns the same angle counter-clockwise, so its tip moves towards $+\mathbf{i}$: its leg along $\mathbf{k}$ (adjacent) is $\cos\Omega t$ and its leg along $\mathbf{i}$ (opposite) is $+\sin\Omega t$.
- $\mathbf{j}_1 = \mathbf{j}$, the axis of the rotation.

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}, \qquad \mathbf{j}_1 = \mathbf{j}, \qquad \mathbf{k}_1 = \sin\Omega t\,\mathbf{i} + \cos\Omega t\,\mathbf{k}$$

The columns of the change-of-basis matrix are these components:

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\Omega t\cos\Omega t - \sin\Omega t\,(-\sin\Omega t) = \cos^2\Omega t + \sin^2\Omega t = 1$$

Each column has unit length ($\cos^2\Omega t + \sin^2\Omega t = 1$) and the first and third columns are orthogonal ($\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1. The origin is fixed ($\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$), $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$ and $\boldsymbol{\alpha}_{10} = \dot\Omega\,\mathbf{j} = \mathbf{0}$ for constant Ω. Check with Poisson's formula: by the chain rule, $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\Omega\sin\Omega t\,\mathbf{i} - \Omega\cos\Omega t\,\mathbf{k} = -\Omega\,\mathbf{k}_1$, and $\boldsymbol{\omega}_{10}\times\mathbf{i}_1 = \Omega\,\mathbf{j}\times\mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.

Kinematics

Fixed curve (parts a to c). The basis $\mathcal{B}_0$ is constant. The height depends on time through $x(t)$, so by the chain rule $\dot y = \frac{dy}{dx}\,\dot x = \sin(x/a)\,\dot x$:

$$\mathbf{v}^P = \dot x\,\mathbf{i} + \sin\frac{x}{a}\,\dot x\,\mathbf{j}, \qquad v = \frac{ds}{dt} = \frac{ds}{dx}\,\dot x = \sqrt{1+\sin^2\left(\frac{x}{a}\right)}\,\dot x, \qquad \mathbf{v}^P = v\,\mathbf{e}_t$$

Differentiating again, product rule on $\sin(x/a)\,\dot x$ with the chain rule $\frac{d}{dt}\sin\frac{x}{a} = \frac{1}{a}\cos\frac{x}{a}\,\dot x$:

$$\mathbf{a}^P = \ddot{x}\,\mathbf{i} + \left(\sin\frac{x}{a}\,\ddot{x} + \frac{1}{a}\cos\frac{x}{a}\,\dot{x}^2\right)\mathbf{j}$$

Rotating curve (parts d to g). In S_1 the vectors $\mathbf{i}_1, \mathbf{j}$ are constant and only the components of $\mathbf{r}_1^P$ change; the same chain and product rules give

$$\mathbf{v}_1^P = \dot{x}_1\,\mathbf{i}_1 + \sin\frac{x_1}{a}\,\dot{x}_1\,\mathbf{j}, \qquad \mathbf{a}_1^P = \ddot{x}_1\,\mathbf{i}_1 + \left(\sin\frac{x_1}{a}\,\ddot{x}_1 + \frac{\dot{x}_1^2}{a}\cos\frac{x_1}{a}\right)\mathbf{j}$$

Projections used below, with $\sqrt{\cdot} \equiv \sqrt{1+\sin^2(x_1/a)}$: $\mathbf{a}_1^P \cdot \mathbf{e}_t = \dfrac{\left(1+\sin^2\frac{x_1}{a}\right)\ddot{x}_1 + \frac{1}{a}\sin\frac{x_1}{a}\cos\frac{x_1}{a}\,\dot{x}_1^2}{\sqrt{\cdot}}$ and $\mathbf{a}_1^P \cdot \mathbf{n} = \dfrac{-\sin\frac{x_1}{a}\,\ddot{x}_1 + \sin\frac{x_1}{a}\,\ddot{x}_1 + \frac{\dot{x}_1^2}{a}\cos\frac{x_1}{a}}{\sqrt{\cdot}} = \dfrac{\dot{x}_1^2\cos(x_1/a)}{a\sqrt{\cdot}}$ (the same with x for the fixed curve).

Forces and Equations of Motion

Real forces. The weight $\mathbf{W} = -mg\,\mathbf{j}$ (conservative, $V = mgy$) and the reaction $\mathbf{N}$ of the smooth curve, perpendicular to $\mathbf{e}_t$, so $\mathbf{N}\cdot\mathbf{v} = 0$. With the projections $\mathbf{j}\cdot\mathbf{e}_t = \sin\alpha = \dfrac{\sin(x/a)}{\sqrt{1+\sin^2(x/a)}}$ and $\mathbf{j}\cdot\mathbf{n} = \cos\alpha = \dfrac{1}{\sqrt{1+\sin^2(x/a)}}$.

Equation of motion, fixed curve.

> **Pedagogical Justification:** projecting Newton's second law $m\mathbf{a}^P = \mathbf{W} + \mathbf{N}$ on $\mathbf{e}_t$ removes $\mathbf{N}$.

$m\,\mathbf{a}^P\cdot\mathbf{e}_t = \mathbf{W}\cdot\mathbf{e}_t$; multiplying both sides by $\sqrt{1+\sin^2(x/a)}$ and dividing by m:

$$\left(1+\sin^2\frac{x}{a}\right)\ddot{x} + \frac{1}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}^2 + g\sin\frac{x}{a} = 0$$

The energy integral of part (a) is the first integral of this equation (work-energy theorem). Check: differentiating $\frac{1}{2}m(1+\sin^2(x/a))\dot{x}^2 + mga(1-\cos(x/a)) = \text{const}$ with the product rule and the chain rule ($\frac{d}{dt}\sin^2\frac{x}{a} = \frac{2}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}$, $\frac{d}{dt}\cos\frac{x}{a} = -\frac{1}{a}\sin\frac{x}{a}\,\dot{x}$) gives

$m\left(1+\sin^2\frac{x}{a}\right)\dot{x}\ddot{x}+\frac{m}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}^3+mg\sin\frac{x}{a}\,\dot{x}=0$, which is $m\dot{x}$ times the equation above.

(d) Inertia forces in the moving frame

> **Pedagogical Justification:** in the non-inertial frame S_1, $m\mathbf{a}_1^P=\mathbf{F}+\mathbf{F}_i$ (Notes Eq. 4.63) with the inertia forces of Notes Eq. 4.62 (Slide 28). The origin is fixed and Ω is constant, so the force of the acceleration of the origin and the Euler force vanish ($\mathbf{F}_A=\mathbf{F}_E=\mathbf{0}$).

1. Centrifugal force. With $\mathbf{j}\times\mathbf{i}_1=-\mathbf{k}_1$, $\mathbf{j}\times\mathbf{j}=\mathbf{0}$ and $\mathbf{j}\times\mathbf{k}_1=\mathbf{i}_1$:

$$\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P=\Omega\,\mathbf{j}\times(x_1\mathbf{i}_1+y\,\mathbf{j})=-\Omega x_1\,\mathbf{k}_1,\qquad \mathbf{F}_{cf}=-m\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P)=-m\,\Omega\,\mathbf{j}\times(-\Omega x_1\mathbf{k}_1)=m\Omega^2x_1\mathbf{i}_1$$

2. Coriolis force:

$$\mathbf{F}_{Cor}=-2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P=-2m(\Omega\mathbf{j})\times(\dot{x}_1\mathbf{i}_1+\dot{y}_1\mathbf{j})=-2m\Omega\dot{x}_1\,(\mathbf{j}\times\mathbf{i}_1)=2m\Omega\dot{x}_1\,\mathbf{k}_1$$

The total inertia force is $\mathbf{F}_i=m\Omega^2x_1\,\mathbf{i}_1+2m\Omega\dot{x}_1\,\mathbf{k}_1$. Its projections: $\mathbf{F}_{cf}\cdot\mathbf{e}_t=\dfrac{m\Omega^2x_1}{\sqrt{\cdot}}$, $\mathbf{F}_{cf}\cdot\mathbf{n}=-\dfrac{m\Omega^2x_1\sin(x_1/a)}{\sqrt{\cdot}}$, and $\mathbf{F}_{Cor}$ is along $\mathbf{k}_1$, normal to the plane of the curve.

Equations of motion in S_1. The reaction of the smooth curve is $\mathbf{N}=N_n\,\mathbf{n}+N_z\,\mathbf{k}_1$ (no tangential component). Projections of $m\mathbf{a}_1^P=\mathbf{W}+\mathbf{N}+\mathbf{F}_{cf}+\mathbf{F}_{Cor}$:

- on $\mathbf{e}_t$ (multiplied by $\sqrt{\cdot}/m$), the tangential projection of the centrifugal force adds Ω^2x_1 to the right-hand side:

$$\left(1+\sin^2\frac{x_1}{a}\right)\ddot{x}_1+\frac{1}{a}\sin\frac{x_1}{a}\cos\frac{x_1}{a}\,\dot{x}_1^2+g\sin\frac{x_1}{a}=\Omega^2x_1$$

- on $\mathbf{n}$: $\dfrac{m\dot{x}_1^2\cos(x_1/a)}{a\sqrt{\cdot}}=N_n-\dfrac{mg}{\sqrt{\cdot}}-\dfrac{m\Omega^2x_1\sin(x_1/a)}{\sqrt{\cdot}}$;
- on $\mathbf{k}_1$: $0=N_z+2m\Omega\dot{x}_1$ ($\mathbf{a}_1^P$ and $\mathbf{W}$ have no $\mathbf{k}_1$ component).

Answers and Checks

Part (a): Reduction of Motion to Integrals

Pedagogical Justification: The frictionless constraint does zero work ($W_N = 0$). Gravity is conservative with potential energy $V(y) = mgy = mga(1 - \cos(x/a))$. Total mechanical energy is constant.

$$E = \frac{1}{2}mv^2 + mgy = \frac{1}{2}mv_0^2 + mg(0) = \frac{1}{2}mv_0^2$$

$$\frac{1}{2}m\left[1 + \sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2 + mga\left(1 - \cos\frac{x}{a}\right) = \frac{1}{2}mv_0^2$$

Multiply by $2/m$:

$$\left[1 + \sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2 = v_0^2 - 2ga\left(1 - \cos\frac{x}{a}\right)$$

Isolate $\dot{x} = \frac{dx}{dt}$:

$$\frac{dx}{dt} = \pm\sqrt{\frac{v_0^2 - 2ga(1 - \cos(x/a))}{1 + \sin^2(x/a)}}$$

Separation of variables and Barrow's rule from $x(0) = 0$ yield the quadrature:

$$\int_0^{x(t)} \sqrt{\frac{1 + \sin^2(x'/a)}{v_0^2 - 2ga(1 - \cos(x'/a))}}\,dx' = \pm t$$

Part (b): Turning Points for $v_0 = \sqrt{ga}$

> **Pedagogical Justification:** Turning points occur where the velocity vanishes ($v = 0 \iff \dot{x} = 0$).

$$v_0^2 - 2ga\left(1 - \cos\frac{x}{a}\right) = 0$$

Substitute $v_0^2 = ga$:

$$ga - 2ga\left(1 - \cos\frac{x}{a}\right) = 0$$

Divide by ga:

$$1 - 2\left(1 - \cos\frac{x}{a}\right) = 0 \implies 2\cos\frac{x}{a} = 1 \implies \cos\frac{x}{a} = \frac{1}{2}$$

In the domain $x/a \in [-\pi, \pi]$:

$$\frac{x}{a} = \pm\frac{\pi}{3} \implies x_{min} = -\frac{\pi}{3}a, \qquad x_{max} = +\frac{\pi}{3}a$$

Part (c): Minimum Velocity v_0 to Reach $x = \pi a$

> **Pedagogical Justification:** At the crest $x = \pi a$, the height is $y(\pi a) = a(1 - \cos\pi) = a(1 - (-1)) = 2a$. For the particle to reach this crest, its kinetic energy at the peak must be non-negative ($T(\pi a) \geq 0$).

$$E \geq V(\pi a) \implies \frac{1}{2}mv_0^2 \geq mg(2a) = 2mga$$

$$v_0^2 \geq 4ga \implies v_{0,min} = \sqrt{4ga} = 2\sqrt{ga}$$

At exactly $v_0 = 2\sqrt{ga}$ the crest is a maximum of V (zero force there), so it is reached only asymptotically ($t \to \infty$); for $v_0 > 2\sqrt{ga}$ it is reached in finite time. $2\sqrt{ga}$ is therefore the infimum of the admissible values.

Part (d): Moving Reference Frame and Inertia Force

The moving frame is $S_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$, attached to the curve and turning with $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$; the position of P in it is $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y(x_1)\mathbf{j}$ and its relative velocity $\mathbf{v}_1^P = \dot{x}_1\mathbf{i}_1 + \sin(x_1/a)\dot{x}_1\mathbf{j}$. The inertia force is

$$\mathbf{F}_i = m\Omega^2 x_1\,\mathbf{i}_1 + 2m\Omega\dot{x}_1\,\mathbf{k}_1$$

(centrifugal plus Coriolis; $\mathbf{F}_A = \mathbf{F}_E = \mathbf{0}$).

Part (e): Quadratures and Reaction Force in the Moving Frame

- Effective potential in S_1:

$$V_{eff}(x_1) = mgy(x_1) - \frac{1}{2}m\Omega^2 x_1^2 = mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2$$

- Energy conservation in S_1 (the curve is fixed and smooth in S_1, so $\mathbf{N}\cdot\mathbf{v}_1^P = 0$; $\mathbf{F}_{Cor}\cdot\mathbf{v}_1^P = 0$ always; gravity is conservative and $V_{cf} = -\frac{1}{2}m\Omega^2x_1^2$ does not depend on time because Ω is constant):

$$E' = \frac{1}{2}m\left[1 + \sin^2(x_1/a)\right]\dot{x}_1^2 + mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2x_1^2 = \text{constant}$$

- **Initial value.** At $x_1 = 0$ the position is $\mathbf{0}$, so $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \mathbf{0}$ and the relative speed equals v_0: $E' = \frac{1}{2}mv_0^2$. Solving for $\dot{x}_1$ and separating variables (Barrow's rule from $x_1(0) = 0$):

$$\dot{x}_1^2 = \frac{v_0^2 - 2ga\left(1 - \cos\frac{x_1}{a}\right) + \Omega^2 x_1^2}{1 + \sin^2(x_1/a)} \implies \pm t = \int_0^{x_1} \sqrt{\frac{1 + \sin^2(\xi/a)}{v_0^2 - 2ga\left(1 - \cos(\xi/a)\right) + \Omega^2\xi^2}}\, d\xi$$

- **Reaction.** $\mathbf{N} = N_n\,\mathbf{n} + N_z\,\mathbf{k}_1$ with the in-plane unit normal $\mathbf{n} = \left(-\sin\frac{x_1}{a}\,\mathbf{i}_1 + \mathbf{j}\right)/\sqrt{1 + \sin^2(x_1/a)}$. Along the normal to the plane, from the $\mathbf{k}_1$ projection: $N_z = -F_{Cor,z} = -2m\Omega\dot{x}_1$. Along $\mathbf{n}$, solving the $\mathbf{n}$ projection for N_n:

$$N_n = \frac{m}{\sqrt{1 + \sin^2(x_1/a)}}\left[g + \Omega^2 x_1 \sin\frac{x_1}{a} + \frac{\dot{x}_1^2}{a}\cos\frac{x_1}{a}\right]$$

with $\dot{x}_1^2$ from the energy integral above.

Part (f): Transcendental Equilibrium Equation

Equilibrium occurs where $\frac{dV_{eff}}{dx_1} = 0$ (relative rest in the tangential equation, $\ddot{x}_1 = \dot{x}_1 = 0$):

$$\frac{dV_{eff}}{dx_1} = mg\sin\left(\frac{x_1}{a}\right) - m\Omega^2 x_1 = 0 \implies g\sin\left(\frac{x_1}{a}\right) - \Omega^2 x_1 = 0$$

Part (g): Case $\Omega = \Omega(t)$

If Ω varies with time, an Euler force $\mathbf{F}_{Euler} = -m\dot{\Omega}\,\mathbf{j} \times \mathbf{r}_1^P = m\dot{\Omega}x_1\,\mathbf{k}_1$ appears; it is normal to the plane of the curve, so $\mathbf{F}_{Euler}\cdot\mathbf{v}_1^P = 0$ and it does no work in S_1 (it is balanced by N_z). The energy is lost for another reason: the centrifugal potential depends explicitly on time, $\frac{\partial V_{cf}}{\partial t} = -m\Omega\dot{\Omega}x_1^2 \neq 0$, so $\frac{dE'}{dt} = -m\Omega\dot{\Omega}x_1^2 \neq 0$. Thus the mechanical energy in S_1 is **not** conserved and the energy equation cannot be used in the same way.

Checks

- **Shape of the potential.** The potential well has width $2\pi a$ and barrier height $2a$ (potential energy $2mga$). Rotation creates a centrifugal destabilizing potential $-\frac{1}{2}m\Omega^2x_1^2$ so that the origin loses stability for $\Omega^2 > g/a$. Off-centre equilibria satisfy $\sin u = \lambda u$ with $u = x_1/a$ and $\lambda = \Omega^2 a/g$; since $\sin u/u$ decreases monotonically from 1 to 0 on $(0, \pi]$, they exist (one pair $\pm u^*$ with $0 < u^* < \pi$) only for $\lambda < 1$, i.e. $\Omega^2 < g/a$, and there they are unstable ($V''_{eff} = (mg/a)(\cos u^* - \lambda) < 0$).

- **Limit $\Omega \to 0$:** $V_{eff} \to mga\,(1-\cos(x_1/a))$, $N_z \to 0$, the quadrature of (e) reduces to that of (a) and the equilibrium equation $g\sin(x_1/a)=0$ leaves only $x_1=0$ inside $(-\pi a, \pi a)$.
- **Orthogonality:** $\mathbf{e}_t \cdot \mathbf{n} = 0$, $|\mathbf{e}_t| = |\mathbf{n}| = 1$ and $\mathbf{e}_t \times \mathbf{n} = \mathbf{k}_1$ (worked out in the change-of-basis section); $[{}_0R_1]$ is orthogonal with $\det = 1$.
- **Numeric check:** a scan of $\sin u = \lambda u$ on $(0,\pi]$ finds one root for $\lambda = 0.5$ ($u^* = 1.8955$) and for 0.9, and none for $\lambda = 1, 1.1, 2$.
- **Dimensions:** $[dy/dx] = [\sin(x/a)] = 1$ and $[ds] = [dx] = \mathrm{m}$; $[g/a] = [\Omega^2] = \mathrm{s}^{-2}$, so $[\lambda] = [\Omega^2 a/g] = 1$; every term of E' is in J ($[mga] = [m\Omega^2 x_1^2] = [mv_0^2] = \mathrm{kg\,m^2 s^{-2}}$); $[N_z] = [m\Omega\dot{x}_1] = \mathrm{N}$; $[g\sin(x/a)] = [\Omega^2 x_1] = \mathrm{m\,s^{-2}}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 22 in its solution keys, so the results are verified by independent derivation (including the numeric scan of the equilibrium equation above).

Source: Problems.pdf